

The multiplicity- m theorem in box form

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Abstract

The chart standard integral over the genuine box $(0, b]^{d'+m}$ with general continuous ξ, η is identified by Fubini with the block state integral of unit 77 (first d' coordinates noncritical, last m the minimal block). Hence, for the last $m = d_0 + 2$ exponents equal to p and the first d' above p , $\int_{(0,b]^{d'+m}} w^H \eta(w) e^{-\beta n w^{2K} + \beta \sqrt{n} w^K \xi(w)} dw \sim \frac{C(\xi, \eta)}{2^{d_0+1}} n^{-p/2} \log^{d_0+1} n$ with $C(\xi, \eta) = \frac{\prod k_i^{-1}}{(d_0+1)!} \int v^{h'} (v^{k'})^{-p} A_{p-1}(\xi(v, 0)) \eta(v, 0) dv$, and the free-energy form $-\log Z = \frac{p}{2} \log n - (d_0 + 1) \log \log n + F_0 + o(1)$. Zero sorry/axiom.

1 The things formalised

```
noncomputable def splitEquiv (d' m : ℕ) : (Fin (d' + m) → ℝ) → (Fin d' → ℝ) × (Fin m → ℝ)
theorem splitEquiv_measurePreserving (b : ℝ) (d' m : ℕ) :
  MeasurePreserving (splitEquiv d' m) (boxMeasure b (d' + m))
    ((boxMeasure b d').prod (boxMeasure b m))
theorem splitEquiv_symm_apply (d' m : ℕ) (v : Fin d' → ℝ) (u : Fin m → ℝ) :
  (splitEquiv d' m).symm (v, u) = Fin.append v u
theorem boxIntegralGen_eq_blockStateIntegral (b N : ℝ) {d' m : ℕ}
  (K H : Fin (d' + m) → ℝ) (hK : (Fin (d' + m) → ℝ) → ℝ)
  (hC : Continuous hK) (hC : Continuous hK) :
  boxIntegralGen b N K H
    = blockStateIntegral b N (fun i => K (Fin.natAdd d' i))
      (fun i => H (Fin.natAdd d' i))
      (fun j => K (Fin.castAdd m j)) (fun j => H (Fin.castAdd m j))
      (fun u v => (Fin.append v u)) (fun u v => (Fin.append v u))
theorem boxIntegralGen_mult_isEquivalent (b p : ℝ) (h : 0 < h) (hb : 0 < b) {d' d : ℕ}
  (K H : Fin (d' + (d + 2)) → ℝ) (hK : (Fin (d' + (d + 2)) → ℝ) → ℝ)
  (hp : (i : Fin (d + 2), ((H (Fin.natAdd d' i)) : ℝ) + 1) / K (Fin.natAdd d' i) = p)
  (hq : (j : Fin d',
    p < ((H (Fin.castAdd (d + 2) j)) : ℝ) + 1) / K (Fin.castAdd (d + 2) j))
  (hC : Continuous hK) (hC : Continuous hK)
  (hpos : (v : Fin d' → ℝ), (j, 0 < v j - v j - b) → 0 < (Fin.append v 0)) :
  (fun n : ℝ => boxIntegralGen b (Real.sqrt n) K H) ~[atTop]
    (fun n : ℝ => blockCoeff b p ... / 2 ^ (d + 1) * (n ^ (-
      (p / 2)) * Real.log n ^ (d + 1)))
theorem boxIntegralGen_mult_freeEnergy ... :
  F : ℝ, Tendsto (fun n : ℝ => -Real.log (boxIntegralGen b (Real.sqrt n) K H)
    - (p / 2 * Real.log n - (d + 1 : ℝ) * Real.log (Real.log n))) atTop ( F)
```

2 Proof strategy

The splitting equivalence is `piCongrLeft` along `finSumFinEquiv` followed by `sumPiEquivProdPi`; both are measure preserving for `pi` measures (`measurePreserving_piCongrLeft`, `measurePreserving_sumPiEquivProdPi`), and its inverse is `Fin.append` (checked coordinatewise by `Fin.addCases` and `piCongrLeft_apply_apply`). `MeasurePreserving.integral_comp'` transports the box integral to the product measure, `integral_prod` iterates it with the noncritical coordinates outermost, and `Fin.prod_univ_add` splits the monomials. The asymptotics are then unit 79 applied to $\xi(v, u), \eta(v, u)$ and the free energy is `tendsto_log_of_isEquivalent_powLog`.

3 Roadblocks and resolutions

- A set `e := splitEquiv d' m` hides the equivalence from `symm_apply_apply` (the goal shows `e.symm ((splitEquiv d' m) w)`); use the explicit name throughout.
- An anonymous `-Real.log _` witness cannot be synthesised; spell out the constant.
- `ring` closing via its `ring_nf` fallback emits a “Try this” note; write `ring_nf` directly.

4 Outcome

The leading term of `thm:TaylorTree` is now available on the genuine chart box for general continuous ξ, η , at every multiplicity, in the sorted coordinate order. Remaining polish: arbitrary coordinate order via the sorting permutation (a general-‘, ‘ permutation lemma), and the paper-facing remainder/expansion theory (Astra targets 6–8).